Cisco PE Device - Routing Protocol Flap Troubleshooting

Node: HSR-PE-07 | **Vendor:** Cisco | **Severity:** Minor | **Alarm Code:** ALARM_9003

Description

The Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Node ID: HSR-PE-07, located at HSR Layout Access, can occur due to the complex interactions between different routing protocols and control plane mechanisms. On Cisco PE devices, the BGP route churn can be triggered by changes in the IGP (OSPF or ISIS) routing table, which can lead to a flap in the BGP routing table. This, in turn, can cause the BGP session to flap, resulting in a loss of adjacency between the PE and CE devices. Additionally, ISIS LSP flooding can occur due to a misconfiguration or a hardware issue, leading to a loss of adjacency between the CE and PE devices. The OSPF adjacency loss can be caused by a variety of factors, including a mismatch in the OSPF configuration between the CE and PE devices, a hardware issue, or a software bug. In order to troubleshoot this issue, it is essential to understand the underlying protocol and control plane mechanisms that are driving the flap.

Pre-requisite Checks

- check: Verify that the redundant link is up and active, description: Ensure that the redundant link is up and active to prevent any impact on the traffic

- check: Verify that the CE devices are configured correctly, description: Ensure that the CE devices are configured correctly to prevent any mismatch in the OSPF configuration

- check: Dispatch remote hands to HSR Layout Access, description: Dispatch remote hands to HSR Layout Access to perform physical checks

Potential Root Causes

Cause	Probability
BGP Route Churn due to IGP Routing Table Changes	High
ISIS LSP Flooding due to Misconfiguration or Hardware Issue	Medium
OSPF Adjacency Loss due to Mismatch in OSPF Configuration or Hardware Issue	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP router identifier 10.0.0.1, local AS number 65000 BGP table version is 10 BGP scanning timer 90 seconds Status codes: s suppressed, d damped, h history, * valid, > best, i - internal, r RIB-failure, S Stale Origincode: i - IGP, e - EGP, ? - incomplete, Description: This command displays the BGP summary, which includes the BGP router identifier, local AS number, and BGP table version. It also displays the BGP scanning timer
```

```
and status codes. This command helps to identify if there are any
issues with the BGP routing table.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 10.0.0.2 1 FULL/DR 00:00:32 10.0.0.2
GigabitEthernet0/1, Description: This command displays the OSPF
neighbor table, which includes the neighbor ID, priority, state, dead
time, and interface. This command helps to identify if there are any
issues with the OSPF adjacency.

Command: show isis database, Expected Output: ISIS LSP Database
(Sequence/LS ID/AGE)
(0x40000001/0x0000.0000.0000.0000.0000.0000.0000.0000.00/
0000.0000.0000.0000.00/0) L 0x80000000 0x0000.0000.0000.0000.00 0x0
(0x40000001/0x0000.0000.0000.0000.0000.0000.0000.0000.00/
0000.0000.0000.0000.00/0) L 0x80000000 0x0000.0000.0000.0000.00 0x0,
Description: This command displays the ISIS LSP database, which
includes the LSP sequence, LS ID, age, and type. This command helps to
identify if there are any issues with the ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Clear BGP Session, Command: clear ip bgp *, Description: This command clears the BGP session, which can help to resolve the BGP route churn issue. Note that this command will clear all BGP sessions, so use with caution.

Step 2: Step: Reset OSPF Neighbor, Command: clear ospf neighbor *, Description: This command resets the OSPF neighbor, which can help to resolve the OSPF adjacency loss issue. Note that this command will reset all OSPF neighbors, so use with caution.

Step 3: Step: Reset ISIS LSP, Command: clear isis database, Description: This command resets the ISIS LSP, which can help to resolve the ISIS LSP flooding issue. Note that this command will reset all ISIS LSPs, so use with caution.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP router identifier
10.0.0.1, local AS number 65000 BGP table version is 10 BGP scanning
timer 90 seconds Status codes: s suppressed, d damped, h history, *
valid, > best, i - internal, r RIB-failure, S Stale Origincode: i -
IGP, e - EGP, ? - incomplete, Description: This command verifies that
the BGP routing table is stable and there are no issues with the BGP
session.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface 10.0.0.2 1 FULL/DR 00:00:32 10.0.0.2
GigabitEthernet0/1, Description: This command verifies that the OSPF
adjacency is stable and there are no issues with the OSPF neighbor.

Command: show isis database, Expected Output: ISIS LSP Database
(Sequence/LS ID/AGE)
(0x40000001/0x0000.0000.0000.0000.0000.0000.0000.0000.00/
0000.0000.0000.0000.00/0) L 0x80000000 0x0000.0000.0000.0000.00 0x0
(0x40000001/0x0000.0000.0000.0000.0000.0000.0000.0000.00/
0000.0000.0000.0000.00/0) L 0x80000000 0x0000.0000.0000.0000.00 0x0,
Description: This command verifies that the ISIS LSP is stable and
there are no issues with the ISIS LSP flooding.
```

Roll-back Procedures

- Step: Restore BGP Session, Command: no clear ip bgp *, Description: This command restores the BGP session, which can help to resolve the BGP route churn issue.

- Step: Restore OSPF Neighbor, Command: no clear ospf neighbor *, Description: This command restores the OSPF neighbor, which can help to resolve the OSPF adjacency loss issue.

- Step: Restore ISIS LSP, Command: no clear isis database, Description: This command restores the ISIS LSP, which can help to resolve the ISIS LSP flooding issue.